Haripriyan Uthayakumar

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Professional Summary

Chemical and biological engineer with expertise in sustainability assessment, process systems engineering, and computational modeling for chemical and manufacturing systems. Experienced in integrating process simulation, material flow analysis, optimization, and machine learning to address challenges in circularity, resource efficiency, supply-chain resilience, and sustainable manufacturing. Lead researcher on an NSF-funded Future Manufacturing project with 6 peer-reviewed publications. Seeking a Ph.D. internship position for **Summer 2027**.

Technical Skills

Sustainability & Systems Engineering: Life Cycle Assessment (LCA), Circular Economy, Material Flow Analysis (MFA), Physical Input-Output Analysis (PIOT), Techno-Economic Analysis (TEA), Waste Valorization & Solvent Recovery, Supply Chain Resilience, Satellite Remote Sensing for GHG Monitoring, Process Systems Engineering

Process Modeling & Optimization: Aspen Plus, Aspen Custom Modeler, SuperPro Designer, DWSIM, MATLAB/Simulink, Process Simulation, Process Network Synthesis, Superstructure Optimization, Design of Experiments (DoE), Gurobi, IPOPT

AI, Machine Learning & Scientific Computing: Python, PyTorch, TensorFlow, JAX, scikit-learn, pandas, NumPy, Hybrid & Physics-Informed Machine Learning, Surrogate Modeling, Bayesian Optimization, Julia, JMP

Generative & Agentic AI: Large Language Models (LLMs), Agentic AI & Multi-Agent Workflows (LangChain, LangGraph), Retrieval-Augmented Generation (RAG), Prompt Engineering, AI-Assisted LCA Automation

Data Analytics & Digital Tools: SQL, MongoDB, Power BI, Automated Data Pipelines, Git, SAS

Professional Skills: Interdisciplinary and academia–industry collaboration, scientific writing and technical presentation, research project leadership, translating models into decision-ready insights for stakeholders

Research Experience

Graduate Research Assistant - SINCS Research Group **Aug 2023 – Present**
Purdue University *West Lafayette, IN*

- Develop process-informed sustainability assessment frameworks that couple Aspen Plus process models with economy-wide material flow analysis to evaluate onshoring and resilience of U.S. active pharmaceutical ingredient (API) supply chains.
- Developed a multi-scale framework integrating chemistry and hybrid mechanistic surrogate models (Aspen + machine learning) for solvent recovery technology selection in sustainable pharmaceutical manufacturing.
- Designed chemistry-driven industrial symbiosis networks to implement circular economy strategies in emerging pharmaceutical industries (presented at AIChE 2025).
- Lead graduate researcher on an **NSF Future Manufacturing Research Grant (FMRG)** for continuous pharmaceutical manufacturing, collaborating with interdisciplinary teams across process systems engineering, pharmaceutical sciences, and industry partners.

Project Scientist **Aug 2022 – Jul 2023**
Research Centre for Atomistic Modeling & Materials Design, ***IIT Madras*** */ PI: Dr. Tarak Patra* *Chennai, India*

- Developed **QRChEM**, a QR-code-encoded CNN deep-learning framework for materials and molecular property prediction and design, published in *ACS Engineering Au*.

Education

Purdue University **May 2025 – May 2028**
Ph.D. in Biological Engineering (Concentration: Computational Science and Engineering) *West Lafayette, IN*

- *Dissertation:* Process-Informed Sustainability Assessment and Machine Learning for Resilient, Circular Continuous Manufacturing of Active Pharmaceutical Ingredients.

Purdue University **Aug 2023 – May 2025**
MS in Biological Engineering (Concentration: Computational Science and Engineering) *West Lafayette, IN*

- *Thesis:* Computer-Aided Design of Frameworks Towards Sustainable Pharmaceutical Manufacturing.

Anna University **Aug 2018 – May 2022**
Bachelor of Technology (B.Tech), Chemical Engineering *Chennai, India*

- *Thesis:* Process network synthesis and Optimization for Net-zero waste in chemical manufacturing networks

Projects

Agentic AI and Satellite-Informed Life Cycle Assessment for Carbon Footprinting of Pharmaceutical API Manufacturing **2026 – Present**

- Architecting a multi-agent LLM workflow (LangGraph + RAG) that automates LCA inventory building by extracting process data from Aspen Plus models, mapping flows to background databases, and drafting ISO 14040/44-aligned interpretations for API synthesis routes.
- Fusing satellite CO_2 and NO_2 observations (OCO-2/3, TROPOMI) with EPA facility-level GHG reporting to benchmark bottom-up emission inventories of U.S. pharmaceutical and upstream chemical plants against top-down estimates.
- Extending the 47+ industry EW-MFA network to Scope 1–3 carbon footprints, linking API onshoring scenarios to regional CO_2 emission impacts.

Mechanistic-Model-Informed EW-MFA Framework for Onshoring and Cascading-Impact Analysis of Acetaminophen API Manufacturing in the United States **2025 – Present**

- Engineered a process-informed Economy-Wide Material Flow Analysis (EW-MFA) framework coupled with Physical Input–Output Tables (PIOTs) to quantify cascading material and resource impacts of pharmaceutical demand shocks across the U.S. chemical manufacturing network.
- Built Aspen Plus models for **47+ process-level industries** spanning key NAICS chemical sectors (via PIOT-Hub), linking unit-operation mass balances to national supply-chain flows.
- Simulated onshoring and production-scaling scenarios from **0.8× to 2.0×** current acetaminophen output, uncovering nonlinear upstream cascades that inform resilient domestic manufacturing strategy (AIChE 2026).

Hybrid Mechanistic Surrogate Framework for Solvent Recovery Technology Selection in Sustainable Pharmaceutical Manufacturing **2023 – 2025**

- Integrated solvent chemistry, Aspen Plus process models, and machine-learning surrogates into a multi-scale framework to screen and rank solvent recovery technologies for pharmaceutical waste streams.
- Replaced repeated rigorous simulations with trained surrogates, accelerating comparison of recovery routes on process performance and sustainability criteria to support waste reduction and solvent reuse decisions (MS thesis).

Hybrid Physics-Informed Machine Learning for Critical Quality Attribute (CQA) Prediction and Real-Time Optimization in Continuous Acetaminophen API Manufacturing **2025 – Present**

- Developed a hybrid mechanistic + data-driven surrogate that **predicts 3 key CQAs** (assay, dissolution, particle size) with **∼97% accuracy**, enabling real-time Quality-by-Design (QbD) process control.
- Coupled the surrogate with Aspen Plus to optimize operating parameters, reducing predicted batch-to-batch CQA variability by **∼35%** and cutting design-iteration time from days to minutes.

Selected Publications

Uthayakumar, H., Peoples, B. K., Olden, J. D., Midway, S., & Singh, S. (2026). Network Connectivity-based Stream Classification for the Conterminous United States. *Scientific Data*, 13, 1070.

Tan, A. S. T., **Uthayakumar, H.**, et al. (2024). Shapley-Shubik Agents within Superstructure-Based Recycling Model: Circular Economy Approaches for Fish Waste Eco-Industrial Park. *Process Integration and Optimization for Sustainability*, 1–15.

Uthayakumar, H., K, R. K., Jain, R., Kumar, R., & Patra, T. K. (2023). QRChEM: A Deep Learning Framework for Materials Property Prediction and Design Using QR Codes. *ACS Engineering Au*, 4(1), 91–98.

Uthayakumar, H., Radhakrishnan, P., Shanmugam, K., & Kushwaha, O. S. (2022). Growth of MWCNTs from *Azadirachta indica* Oil for Optimization of Chromium(VI) Removal Efficiency using Machine Learning Approach. *Environmental Science and Pollution Research*, 29, 34841–34860.

Kushwaha, O. S., **Uthayakumar, H.**, & Kumaresan, K. (2022). Modeling of Carbon Dioxide Fixation by Microalgae using Hybrid Artificial Intelligence (AI) and Fuzzy Logic (FL) Methods. *Environmental Science and Pollution Research*, 29, 20754–20772.

Conference Presentations

Haripriyan Uthayakumar, Shweta Singh, "Mechanistic-Models-Informed EW-MFA Framework for Onshoring and Cascading-Impact Analysis of Pharmaceutical Manufacturing in the United States," *AIChE Annual Meeting 2026*, Minneapolis Convention Center, Minneapolis, MN, USA.

Haripriyan Uthayakumar, Shweta Singh, "Chemistry-Driven Industrial Symbiosis for Implementing Circular Economy in the Emerging Pharmaceutical Industries," *AIChE Annual Meeting 2025*, Marriott Boston Copley, Boston, MA, USA.